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CSE428

Section 03

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Lecture 5

① a)

$$p = (k-1)/2 \\ = (3-1)/2 = 1$$

b)

$$1/16 (6 \times 1 + 2 \times 2 + 1 \times 1 + 2 \times 2 + 0 \times 4 + 2 \times 2 + 1 \times 1 \\ + 9 \times 2 + 9 \times 1) = 2.94 = 3$$

$$1/16 (2 \times 1 + 1 \times 2 + 9 \times 1 + 0 \times 2 + 2 \times 4 + 4 \times 2 + 9 \times 1 \\ + 9 \times 2 + 5 \times 1) = 3.8 = 4$$

$$1/16 (2 \times 1 + 0 \times 2 + 2 \times 1 + 1 \times 2 + 9 \times 4 + 9 \times 2 \\ + 2 \times 1 + 0 \times 2 + 1 \times 1) = 3.9 = 4$$

$$1/16 (0 \times 1 + 2 \times 2 + 4 \times 1 + 9 \times 2 + 9 \times 4 + 5 \times 2 \\ + 0 \times 1 + 1 \times 2 + 2 \times 1) = 4.75 = 5$$

output image :

$$\begin{bmatrix} 3 & 4 \\ 4 & 5 \end{bmatrix}$$

$$c) g_{mask} = \begin{bmatrix} 0 & 2 \\ 9 & 9 \end{bmatrix} - \begin{bmatrix} 3 & 4 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} -3 & -2 \\ 5 & 4 \end{bmatrix}$$

$$d) g = \begin{bmatrix} 0 & 2 \\ 9 & 9 \end{bmatrix} + \begin{bmatrix} -3 & -2 \\ 5 & 4 \end{bmatrix} = \begin{bmatrix} -3 & 0 \\ 14 & 13 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 14 & 13 \end{bmatrix}$$

$$\begin{aligned}
 2) \quad g &= \begin{bmatrix} 0 & 2 \\ 9 & 9 \end{bmatrix} + 2 \begin{bmatrix} -3 & -2 \\ 5 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 9 & 9 \end{bmatrix} + \begin{bmatrix} -6 & -4 \\ 10 & 8 \end{bmatrix} \\
 &= \begin{bmatrix} -6 & -2 \\ 19 & 17 \end{bmatrix}
 \end{aligned}$$

② Sobel vertical mask:

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

③

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

④

a) salt and pepper noise

b) The median filter is a non-linear filter that replaces each pixel with the median of its neighborhood. It effectively removes salt and pepper noise while preserving sharp edges better than linear smoothing filters.

c)

$$L = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

d) Figure 3, containing high frequency details from figure 2, can sharpen the original image using Laplacian sharpening. Adding these details back increases edge contrast, making boundaries appear sharper and more defined.

e) Figure 4, the blurred image is used in unsharp masking by subtracting it from the original to create an edge mask. Adding this mask back to the original, scaled by a factor k , produces the sharpened image.

Lecture 6

(1) Image 1: $N=M=3$

2D DFT of a 3×3 image,

$$F(u, v) = \sum_{x=0}^2 \sum_{y=0}^2 f(x, y) e^{-j2\pi(\frac{ux}{N} + \frac{vy}{N})}$$

$$F(0, 0) = \sum \sum 1 = 9$$

$$|F_1| = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Acts as low pass filter.

Image 2:

$$\text{For } u=0, v=0; F(0, 0) = 3(-1+0+1) = 0$$

$$u=0, v=1; F(0, 1) = -4.5 - j2.59$$

$$|F(0, 1)| = \sqrt{(4.5)^2 + (2.59)^2} = 5.196$$

$$u=0, v=2; |F(0, 2)| = \sqrt{(-4.5)^2 + (2.59)^2} = 5.196$$

$$|F_2| = \begin{bmatrix} 0 & 5.196 & 5.196 \\ 5.196 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

~~Acts as high pass filter that can detect vertical edge.~~

This maps directly to spatial image processing techniques like edge detection, specially, this matrix is identical to the core of horizontal gradient operator (present in Sobel filter) used to detect vertical edges.

(2) and A

(3) ~~A~~ B

(4) ~~A~~ B

(5) ocean $\rightarrow$ Bottom DFT

zebra $\rightarrow$ Top DFT

zebra stripes cause horizontal frequency spread due to vertical patterns, while the ocean horizon creates a vertical line in the DFT due to horizontal structure.

(6)

Image 1: High-pass filter.



It blocks DC component and low frequencies and passes high frequency which represent the rapid intensity changes.

Image 2: Low pass filter.

It passes DC and low frequency and blocks high frequency.

